



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

RI2003-1 – MICRO AERIAL ROBOTS
(For RI).

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. As per the drone categories in India weight of the Large category drone should be _____
A) 2kg to 25kg B) 25kg to 150kg
C) 150kg or more D) Less than 100kg
2. Which one of the following UAV can perfectly execute the mission in between the altitude of 5000m to 15,000m
A) MALE UAV B) HALE UAV
C) TACTICAL UAV D) Close Range UAV
3. What is the primary purpose of wing flaps?
A) To increase thrust B) To reduce engine noise
C) To increase lift D) To control fuel consumption
4. Which aerodynamic force opposes thrust?
A) Drag B) Lift
C) Weight D) Gravity
5. What is the term for the point where all weight of the aircraft is considered to be concentrated?
A) Center of Thrust B) Center of Lift
C) Center of Gravity D) Center of Pressure
6. Which of the following controls pitch movement of an aircraft?
A) Ailerons B) Elevators
C) Rudder D) Flaps
7. What does 'Endurance' of an aircraft measure?
A) Maximum distance traveled B) Maximum time in the air with given fuel
C) Maximum altitude attained D) Maximum thrust produced
8. Which device can reduce the glide ratio during landing to allow shorter runway requirements?
A) Ailerons B) Rudder
C) Flaps D) Propeller
9. What is required for dynamic stability in a system?
A) Equal distribution of forces B) Forces proportional to angular velocity
C) Large tail surface area D) Absence of artificial feedback
10. What does a negative slope in the pitching moment vs. lift coefficient curve indicate?
A) Instability B) Neutral stability
C) Positive static stability D) Static instability
11. What is the primary principle behind solar cells?
A) Electrolysis B) Photovoltaic effect
C) Joule heating D) Magnetic induction

12. What is the primary working fluid for fuel cells in UAVs?
 A) Diesel B) Hydrogen
 C) Jet fuel D) Kerosene
13. Which motor type is more efficient for UAVs due to less friction?
 A) Canned motor B) Brushless motor
 C) DC motor D) AC motor
14. What is the main advantage of brushless motors over canned motors?
 A) Higher friction B) More parts to wear out
 C) Increased efficiency D) Lower torque
15. What is the key difference between unidirectional (UD) and bidirectional (BD) skin fabrics?
 A) Material thickness B) Color of the fabric
 C) Fiber alignment and strength direction D) Core compatibility
16. What happens to the lift during a turn without increasing the total lift?
 A) The UAV will gain altitude B) The UAV will stall
 C) The UAV will lose altitude D) The UAV will maintain level flight
17. What is the purpose of the uplink in the data-link system of an MPCS?
 A) To receive sensor data from the UAV B) To transmit commands to the UAV and its sensors
 C) To record flight path data D) To monitor environmental conditions
18. Which of the following is a characteristic of passive sensors used in reconnaissance payloads?
 A) They emit their own energy to detect targets B) They rely on external energy sources, such as sunlight or heat
 C) They actively scan and transmit radar waves D) They jam enemy communication signals
19. Which of the following is NOT a desirable attribute of a UAV data link system?
 A) Worldwide frequency availability B) Low probability of intercept
 C) High power consumption D) Anti-jamming capability
20. What is the main drawback of increasing transmitter power to enhance data-link margin?
 A) It is not effective against jamming B) It reduces communication range
 C) It requires directional antennas D) It does not affect signal strength

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Classify UAVs based on Altitude and Rotor.	8	L*1	1	1
	b) Discuss the flapping wing mechanism with a sketch.	8	L2	2	1
2.	a) Classify various UAV systems and briefly explain the categories: very small, small, medium, and large UAVs.	8	L2	1	1
	b) Derive the range equation for propeller-driven aircraft.	8	L2	2	1
3.	a) Explain the following terms i. Aircraft polar ii. Total Air Vehicle drag	8	L2	1	1
	b) Explain the following performance characteristics or parameters of an air vehicle. i. Climbing Flight ii. Gliding Flight	8	L2	2	1
Unit – II					
4.	a) With a block diagram explain each component of the autopilot control system.	8	L2	3	1
	b) Explain the principle of working of the following with a neat sketch (i) Four-cycle engine (ii) Gas Turbine Engine	8	L2	4	1

- | | | | | | | |
|----|----|---|---|----|---|---|
| 5. | a) | Short note on overall modes of operation. | 8 | L2 | 3 | 1 |
| | b) | Describe the dynamic loads experienced by UAVs during flight. How do maneuver loads influence structural requirements? | 8 | L2 | 4 | 1 |
| 6. | a) | Derive the equation for thrust generation. | 8 | L2 | 3 | 1 |
| | b) | Explain the concept of sandwich construction in UAV structures and discuss the materials used for skins, cores, and bonding resins. | 8 | L2 | 4 | 1 |

Unit – III

- | | | | | | | |
|----|----|---|---|----|---|---|
| 7. | a) | What is the role of the Mission Planning and Control Station (MPCS) in UAV operations, and how does it manage flight and payload functions? | 8 | L2 | 5 | 1 |
| | b) | How does the physical configuration of the MPCS impact its operational efficiency and portability in different mission environments? | 8 | L2 | 5 | 1 |
| 8. | a) | Describe the essential components of a UAV data link system and their respective functions during mission operations. | 8 | L2 | 5 | 1 |
| | b) | What are the desirable attributes of a data link system for UAVs, and why are they crucial for mission success in both peacetime and wartime scenarios? | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
